

Automated Detection of Policy Frames in German News Headlines

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Abstract

In an information-dense media environment, complex issues are often compressed into a few attention-grabbing words that also frame the event. This work asks whether article frames survive in headlines, whether they can be classified automatically, and whether this reveals differences across German media. An annotation study of 210 headlines finds LLMs more precise and consistent than untrained human annotators, though not consistently more accurate. Using SemEval-2023, the Media Frames Corpus (MFC), and a Chinese News Framing Dataset (CHN), LLM annotation produced 3,010 viable frame labels across 2,496 headlines. A fine-tuned ModernGBERT reaches micro-F1 0.716 (macro-F1 0.532), transferring to German at 0.380–0.420 micro-F1. Evaluation against an independently human-annotated set of 70 Tagesschau headlines yields a similar pattern with micro-F1 of 0.588 and a macro-F1 of 0.379 indicating the model learns headline-level signals on frequent labels. However, sparse labels, reference filtering, and differences between source corpora limit generalization.

1 Problem Outline and Objective

1.1 Describing the Problem

Through their selection and weighting of topics, news media shape the public agenda, reflecting political and social perspectives and audience expectations. What matters is not only which events are reported but also how they are framed which is the central concern of frame analysis.

Frames are interpretive structures that highlight certain aspects of an issue, suggesting problem definitions, causal attributions, evaluations, or courses of action (Entman, 1993). A topic denotes what a report is about while a frame describes the perspective under which it appears, e.g., a traffic regulation portrayed as an economic burden, a health benefit, or a mobility-justice issue. Frames are not equivalent to positive/negative evaluation or to an outlet’s political stance. Different outlets can share a frame,

and one outlet can offer several perspectives on one event (Entman, 1993; Otmakhova et al., 2024).

Framing need not be deliberate manipulation but can be used to promote particular interpretations, as political actors often recast news into preferred narratives. One example, per CORRECTIV (July 2026), is the AfD tool "Alternita Studio," said to prioritize news by expected reach and generate posts with a predetermined slant. According to the AfD’s lawyers it was still in development, with a planned 2027 launch (Peters et al., 2026). This illustrates why framing deserves study alongside the attention an event receives.

Without systematic comparison, users often struggle to see differences between media portrayals. Ground News addresses this by aggregating coverage, highlighting differences, and revealing "blind spots" where coverage skews to one side of the political spectrum. Its bias ratings apply to outlets and are rooted in the U.S. context (Ground News, 2026). A complementary frame analysis captures differences at the level of individual reports not evident from source-level classification alone. Moreover, in a crowded digital environment headlines condense complex issues into a few words meant to grab attention and orient the reader, and are often read independently of the full text. A headline alone can thus convey an initial framing, as elaborated in the theoretical background.

1.2 Objective and Project Contribution

We develop a Ground-News-inspired application for German headlines that automatically detects generic frames and shows their distribution across comparable reports, combining topical grouping with frame analysis. A suitable model needs headline-level training labels, but available corpora target full texts mainly. We investigate how to derive a training resource for much shorter headlines, combining transformer fine-tuning with a Ground-News-style prototype comparing current German outlets. Three research questions guide this work:

- **RQ1:** To what extent are the policy frames annotated at the full-text level already recognizable in news headlines? This question is addressed through an annotation study and a comparison of full-text and headline labels. Agreement with the article reference is distinguished here from an independent assessment of headline accuracy.
- **RQ2:** How reliably can policy frames recognizable at the headline level be automatically classified as a multi-label task? We address this question through headline annotation and model training. As a sub-question, we examine whether training data from other source languages can also be used to detect frames in German news headlines.
- **RQ3:** Can such a model be used to make differences in headline framing among German media visible within the same topics or events? The Ground-News-inspired prototype and its qualitative analysis serve to address this question. A comparison within the same event requires tighter grouping than mere assignment to a shared topic.

Along answering those questions, we aim at four distinct project contributions: 1) Our methodological contribution consists of examining which article labels are recovered at the headline level, together with a unified procedure for automatic headline annotation to generate a training dataset. 2) As a contribution regarding the data basis, we create a conservative pseudo-label dataset for selected data from SemEval-2023, MFC, and CHN, along with a small, jointly agreed-upon gold standard for 70 Tagesschau headlines, which enables a direct evaluation against human headline annotations and supplements the larger pseudo-label set with a limited manual reference. 3) Our technical contribution is a German-language transformer model for multi-label frame detection. 4) Our application-related contribution lies in the implementation of a prototype that groups news headlines from different outlets and compares their predicted frames.

2 Theoretical Background

2.1 Framing and Media Frames

According to [Entman \(1993\)](#), frames select certain aspects of a perceived reality and make them more salient in a communicating text. In doing so, they suggest particular problem definitions, causal attributions, evaluations, or courses of action. For our analysis, what matters are such interpretive offerings as they are recognizable in the text itself. A

single topic can be portrayed from several angles at once. Frame assignments are accordingly not mutually exclusive in principle.

[Pentzold and Fraas \(2023\)](#) describe media frames as adaptive networks of meaning whose specific form depends on the given contextual configuration. From this perspective, we do not understand frames as rigid units that are always expressed in the same way. For our operationalization, this supports allowing multiple assignments and not reducing a headline in advance to exactly one perspective. The decision to treat this as a multi-label task is our own methodological implementation, however, and not a direct requirement of this theoretical approach.

[Otmakhova et al. \(2024\)](#) further note the need to distinguish between frames and topics. In our application, we accordingly treat the grouping of events and the detection of frames as separate tasks.

2.2 Generic Policy Frames

Boydston et al.’s Policy Frames Codebook ([Boydston et al., 2018, 2020](#)) offers a cross-topic frame-annotation approach for U.S. news ([Card et al., 2015](#)), empirically supported for other languages/topics in principle, e.g., SemEval-2023 Task 3 adopts its 14 content frames for multilingual classification ([Piskorski et al., 2023](#)) though not guaranteeing interchangeable cultural/media contexts. [Daffara et al. \(2025\)](#) report broad applicability for Brazilian news alongside rare, very general categories. The scheme is well established but doesn’t guarantee uniform performance across languages or topics. [Mahmoud and Nakov \(2023\)](#) further examine how language order during multilingual fine-tuning affects SemEval-2023 results, motivating further exploration though not directly applicable here.

2.3 News Headlines as the Unit of Analysis

Headlines influence both news selection and interpretation. [Robertson et al. \(2023\)](#) show via randomized Upworthy tests that negative words raise click probability. [Ecker et al. \(2014\)](#) show misleading headlines affect interpretation and recall, so headlines direct attention and preset interpretation. Their brevity poses challenges: only a few words carry the frame, and meaning can depend on names, prior knowledge, and the event itself.

[Liu et al. \(2019\)](#) show individual frames can be classified from U.S. gun-violence headlines. [Akyürek et al. \(2020\)](#) extend this to multilingual multi-label classification and limited-resource transfer. Both support headlines as the unit of anal-

ysis, though it remains open how well the article categories used transfer to German headlines.

We first assume headlines convey a subset of the article’s perspectives, with fewer frame assignments than in the full text tested in the annotation study. And second that headline frames mostly also occur in the article, though not necessarily its dominant frame. Frames absent from the full text are not expected in the headline.

2.4 Methodological Approach

We first conduct an annotation study to examine which article-level frames humans and LLMs identify in headlines. Its findings inform the automated construction of a headline training corpus. Drawing on the principle of weak supervision (Ratner et al., 2017; Hirsbrunner, 2024), we combine potentially imperfect annotation sources through majority voting and filter their decisions against article-level references. The resulting pseudo-labels serve as fixed training targets for ModernGBERT (Wunderle et al., 2025). Manual review of discarded assignments and a separate human-annotated headline test set provide complementary assessments of this procedure.

We deliberately decide against a purely few-shot setup for this final step, since that would be both computationally and financially expensive for an application like Ground News in continuous use. For a later, direct evaluation of the application, we additionally create a small, human-annotated headline reference from Tagesschau news.

3 Data and Annotation Study

3.1 SemEval-2023 Task 3 as the Starting Dataset

SemEval-2023 Task 3 examines genre, framing, and persuasion in multilingual online news (Piskorski et al., 2023), assigning articles one or more of 14 generic frame categories. The official German set has 132 training and 45 dev articles. Since reducing to headline level loses information, we added the English, French, Italian, and Polish SemEval-2023 subsets (Piskorski et al., 2023). Even so, the resulting volume proved too small for fine-tuning after cleaning, so we also drew on an MFC excerpt from the FrameNews-PT repository’s stratified sample (6,214 articles (Card et al., 2015; Daffara et al., 2025)) and 312 Chinese articles from the Chinese News Framing Dataset (Cook et al., 2025a,b). Due to data volume, MFC headlines were translated using DeepL, while Google Translate was used for the other source corpora. We accepted the resulting translation errors and cultural

differences (word choice, ambiguity, emphasis) as unavoidable for this approach.

3.2 Annotation Study

This preliminary study provides the starting point for automated corpus construction. We compare the frames identified by humans and LLMs, agreement among annotators, and overlap with article-level reference labels. No independent headline gold standard is available for this preliminary study.

We draw a language-weighted random sample (15% rate, seed 42) from SemEval-2023’s German, English, French, Italian, and Polish subsets (Piskorski et al., 2023). It comprises 210 headlines, including 77 originally English, 32 French, 27 German, 45 Italian, and 29 Polish. The corresponding articles carry a total of 850 positive frame assignments.

Three human annotators, along with ChatGPT 5.6 Luna, ChatGPT 5.6 Sol, Claude Sonnet 5, and Gemini 3.6 Flash, work on the same German-language (or German-translated) headlines. All receive the guidelines from the Policy Frames Codebook (Boydston et al., 2020). After a joint introduction, the human annotators work independently of one another. A primary frame is marked with 2, other applicable frames with 1, and non-applicable frames with 0. If no frame applies, the prompt explicitly permits a complete null annotation. The four LLM configurations receive the headlines together with the annotation guidelines and the same prompt shown in Appendix A (Prompt 1), and are run separately.

3.3 Evaluation Procedure

We compute nominal and ordinal Krippendorff’s alpha as well as Fleiss’ kappa for the frame assignments. To compare against SemEval, we additionally determine micro- and macro-F1, micro-precision, micro-recall, sample-F1, and exact match for each annotator or LLM. We then average these values within the two groups.

Precision denotes the share of assigned headline labels that are also contained in the article reference. Recall indicates the share of article reference labels recovered at the headline level. Both measures thus capture overlap with the reference rather than an independently confirmed measure of headline accuracy. However, since we generally assume that not all frames annotated in the full texts are also present at the headline level, precision is more important than recall for this evaluation.

Metric	Human	LLMs
Alpha nominal (0/1/2)	0.235	0.371
Alpha ordinal (0/1/2)	0.330	0.466
Alpha nominal (binary)	0.314	0.458
Fleiss-Kappa (binary)	0.313	0.458
Micro-Precision	0.558	0.699
Micro-Recall	0.348	0.279
Micro-F1	0.425	0.395
Macro-F1	0.377	0.343
Exact Match	0.022	0.042

Table 1: Annotation study with 210 titles. Group means are unweighted.

3.4 Results

LLM configurations agree with each other more than untrained human annotators (with one round of annotation) do. Combining primary and additional frames raises nominal alpha in both groups (Table 1). Even after binarization, the LLMs’ alpha (0.458) stays well below full agreement, and this can’t be explained by ChatGPT 5.6 Sol and Luna sharing a model family. Excluding Sol barely changes it (from 0.458 to 0.457), with no especially high pairwise agreement between the two ChatGPT configurations.

Human annotators achieve higher average micro-F1 and recall, while LLMs show higher precision against the article-level references. Among the LLMs, micro-F1 ranges from 0.337 to 0.470. Luna achieves the highest precision at 0.749, while Gemini has the highest micro-F1 at 0.470. These results support using LLMs when reference-based precision is the priority, but do not show that they generally outperform human annotators.

Overlap with article references is greatest for

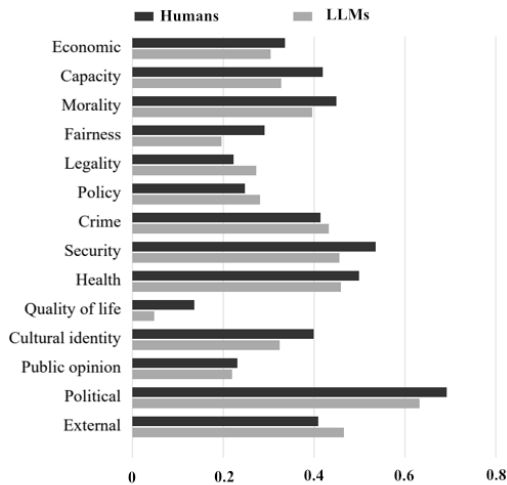


Figure 1: Average F1 per frame compared to SemEval in the annotation study. Our own analysis using abbreviated category names

Political and smallest for Quality of Life (Figure 1). Humans get the higher mean F1 for ten of 14 frames, though binary agreement is very low for Public Opinion (0.005), Quality of Life (0.014), and Policy Prescription and Evaluation (0.076); LLMs likewise show low agreement for Quality of Life (0.090).

By source language, both groups achieve their highest mean micro-F1 values for English headlines, at 0.457 for the humans and 0.443 for the LLMs. The lowest values occur for Polish headlines, at 0.367 and 0.322, respectively. For German headlines, the values are 0.435 and 0.352. Since the language groups comprise only 27 to 77 articles each, these differences primarily describe the given sample. They do not permit a causal explanation in terms of language or culture.

3.5 Implications for Corpus Construction

The annotation study supports the idea that headlines convey only a subset of article perspectives, though low recovery may reflect missing headline signals or overlooked frames. The LLMs’ higher precision and internal agreement make them the more relevant source for automated corpus construction, justifying a conservative labeling approach, without implying human/machine equivalence or full label correctness.

For modeling, we combine primary and additional frames into a single binary multi-label target variable. For each of the 14 content frames, the model is meant to predict its presence at the headline level; multiple frames can be present simultaneously. The higher agreement after binarization supports this simplification. The original distinction between primary and additional frames can be retained as exploratory supplementary information but should not be learned as a rank order. Headline annotations and SemEval full-text labels are treated as separate label sets. The article references serve for comparison and for reference-based selection of training signals, but do not constitute an unmodified headline gold standard.

The LLMs’ higher reference precision motivates a consensus rule for combining annotations, explicitly a precision-oriented pseudo-labeling strategy accepting incompleteness given low recall. An unassigned label thus means selection conditions weren’t met, not confirmed absence. Luna’s higher precision and practical access constraints informed our decision to use the providers’ base models for headline annotation.

4 Construction of the Training Data

4.1 Automated Annotation and Data Selection

SemEval and CHN are annotated by GPT-5.6 Luna, Claude Sonnet 5, and Gemini 3.6 Flash; for MFC, corpus size and token limits require substituting GPT-5.6 Luna, Claude Haiku 4.5, and Mistral GLM-5.2. Using weaker models here may have introduced unintended disparities in the resulting annotations, which complicates comparability across the LLMs. However, this was accepted in favor of a larger dataset.

The MFC prompt was adapted for block-wise processing, but both prompt versions (Appendix A, Prompt 2 and 3) require codebook-compliant annotation, preserving line count, order, metadata, and column structure, using only values 0/1/2, at most one primary frame per headline, and a null vector when no frame applies. The values 0/1/2 were, once again, binarized.

Headline annotation uses a two-stage procedure: majority voting, then article matching. A label is adopted only if at least two of three LLMs assign it. Consensus labels are then matched against full-text labels, assuming a headline reflects selected article perspectives, so results remain pseudo-labels. The significance of this two-stage filter is illustrated by the headline "Pilgrim Condemns Media" ("Pilger verurteilt Medien"), to which two language models assigned the frame "Crime and Punishment." Since this frame is not present in the full-text annotation, it is correctly removed from the final annotation for lack of plausibility.

4.2 Manual Review of Removed Labels

All 446 removed SemEval frame assignments across 377 headlines were manually reviewed. Of these, 339 assignments (76.0%) were confirmed as genuinely incorrect, and 107 (24.0%) were considered plausible in principle. This review thus supports many of the filtering decisions, but also shows that plausible headline frames can be lost in the process. Plausibility here is not equivalent to conclusively confirmed correctness, and is often attributable to semantic compression. Because only removed labels were reviewed, neither the precision of the entire dataset nor the filter's actual net benefit can be determined from this check alone.

4.3 Scope and Selection Effect

Of the 7,654 positive consensus assignments made by the LLMs, 3,010 remain after matching against the original labels. This reference filter thus discards 60.7% of the language models' assignments.

Corpus	Title	Label consensus	Final labels	Pos. titles
CHN	312	680	467	293
MFC	6,214	5,270	1,285	1,285
SemEval	1,401	1,704	1,258	918
Total	7,927	7,654	3,010	2,496

Table 2: Title-frame pair comparison of pre normalization training data. Multi-label assignments possible.

At least one positive label is retained for 2,496 of the 7,927 headlines overall. The corpora differ substantially in this respect: 93.9% of CHN headlines remain positively annotated, compared with 65.5% for SemEval and only 20.7% for MFC (Table 2).

The MFC subset provides only one main-frame reference per article. This frame need not appear in the headline, while other frames expressed in the headline may be absent from the available reference. Thus, at most one positive label can survive the reference filter for each MFC headline. The removal rate of 75.6% hence cannot be interpreted directly as evidence of poorer headline annotation. Overall, Policy Prescription and Evaluation (549) and Legality (528) are most frequent. Quality of Life (31), Cultural Identity (40), Public Opinion (43), and Fairness and Equality (53) are rare, and 113/130 Capacity and Resources assignments come from MFC alone so the model may learn topic/corpus differences alongside frame features.

4.4 Normalization and Residual Category

Cleaning standardizes Unicode and whitespace, merges identical headlines with matching labels, and drops rows with conflicting labels. Combining corpora reduces 7,927 headlines to 7,776 (311 CHN, 6,082 MFC, 1,383 SemEval); 2,492 headlines carry at least one positive frame, 5,284 remain unlabeled.

The training notebook adds the technical residual category "other." It is set to 1 exactly when all 14 content frames are 0. This category differs from the codebook's "other," which denotes a frame signal present outside the 14 dimensions (Boydston et al., 2020). In our training data, "other" can also arise from a lack of consensus or from a removed label. We therefore refer to this category below as other/null case.

The residual category is meant to allow the model to abstain from assigning a substantive frame. Many MFC null cases stem from its restricted main-frame reference and are excluded to avoid dominating training. Null cases are retained only from GER, SemEval, and CHN. The resulting training set comprises 2,590 news headlines:

177 from GER, 819 from the remaining European SemEval subsets, 311 from CHN, and 1,283 from MFC. The 98 null cases are distributed across GER (80) and CHN (18). This selection changes the distribution of target categories. Because "other" is predicted via its own sigmoid output, it remains technically possible to predict it simultaneously with content frames. This means not only that a frameless headline can be assigned, but, in theory, also a frame outside the 14 MFC categories.

5 Fine Tuning and Evaluation

5.1 Model Selection and Training

We use the German-language checkpoint LSX-UniWue/ModernGBERT_134M (Wunderle et al., 2025) with a classification head for 15 binary outputs. Headlines are truncated to a maximum of 128 tokens and padded to this length as needed. The sigmoid outputs allow multiple frames to be predicted simultaneously. Binary cross-entropy serves as the loss function. No rank order among the detected frames is learned.

Each training run starts from the pretrained model. We use a learning rate of 0.00001, a batch size of 16, weight decay of 0.05, ten epochs, and seed 42. Evaluation on the dev set is performed after each epoch. For testing, we select the checkpoint with the highest dev macro-F1. The decision threshold is 0.5. No class weighting is applied. The ten documented training runs take between 148 and 800 seconds each, excluding separate data preparation.

5.2 Two Evaluation Scenarios

We first test dataset combinations, splitting each corpus mix via iterative multi-label stratification (mirroring SemEval-2023 Task 3’s ratio) into approximately 80% training, 12% dev, and 8% test data. Because of the stratified label distribution, the actual sizes may deviate from these figures. These tests describe predictive performance within the given corpus mix. Since the test sets differ between runs, no isolated effect of additional training data can yet be derived at this stage.

The transfer experiments use four training mixtures that exclude the original German headlines. For each mixture, roughly 80% of the data is selected for training, with the remainder left unused. GER is split, using the same stratification and seed 42, into 106 dev and 71 test headlines. These German subsets are identical across all four variants. The experiments thus examine transfer without German training examples, while still using German data to select the model checkpoint.

We report micro-F1, macro-F1 (unweighted per-class mean), and exact match. The values labeled precision and recall in the notebook outputs are macro-averages. Undefined class-level values are set to zero in the implementation. An F1 of zero can therefore also occur when a category has no positive test examples.

6 Classification Results

6.1 Regular Tests

GER alone reaches micro-F1 0.350, macro-F1 0.067; only Health and Safety (0.400) and other/null case (0.600) score positively with just 141 training/14 test headlines, too small to assess broad category detection. The full corpus mix gets the highest macro-F1 (0.532), narrowly ahead of EU+GER+MFC (0.529), which instead leads on micro-F1 (0.762) and exact match (0.647); Complete scores 0.716/0.571 on these. So more data doesn’t win every metric, and differing test populations prevent direct performance ranking.

For the Complete mix, Fairness and Equality, Quality of Life, and Public Opinion remain at F1 = 0. Cultural Identity reaches 0.500 and other/null case 0.200. These values depend on the sample and the chosen threshold. Subsequently removing the null classes would therefore not represent a directly comparable improvement, since it would change both the target space and the denominator of the macro-F1. We therefore proceed with a comparison using a fixed GER test set.

6.2 Transfer to German Headlines

Training w/o GER	N	Micro-F1	Macro-F1
EU	652	0.380	0.209
EU + CHN	894	0.420	0.235
EU + MFC	1,676	0.394	0.232
EU + CHN + MFC	1,925	0.408	0.256

Table 3: Transfer without GER training, using identical German dev and test sets of 106 and 71 titles, respectively. *N* denotes number of training examples.

Relative to EU alone, all extensions improve both micro- and macro-F1 9 (Table 3); EU+CHN gets the highest micro-F1 (0.420), and the full mixture minus GER the highest macro-F1 (0.256), though training sets are redrawn per mixture, so these remain descriptive findings.

The full transfer mixture achieves positive F1 values for seven categories. Eight categories, including other/null case, remain at zero. Health and Safety reaches 0.824, Security and Defense 0.692, and Morality 0.667. Legality and Policy Prescription, by contrast, remain at zero despite being

especially frequent in the overall dataset, showing abundant source examples don’t guarantee reliable German-headline detection.

7 Application and Discussion

The ‘complete’ corpus mix model performs best in the regular tests, so we assess its practical applicability on current German headlines; it was uploaded to Hugging Face as nameless-0/GER_frame-detect_mgbert134M.

7.1 Human Tagesschau Gold Standard

We collected 81 Tagesschau headlines via RSS on September 17, 2026. After deduplication, 70 remained. Three people binarily annotated these across 15 categories. Before discussion, all three agreed fully on only three headlines (4.3%). The mean of 15 category-wise Fleiss’ kappa values is 0.281, pairwise mean Jaccard values range 0.294–0.412 and highest kappas are Health and Safety (0.685) and Security and Defense (0.653).

For the gold standard, all headlines without unanimous annotation were discussed jointly, resulting in a total of 64 positive content frames assigned across 50 headlines. Of these, 37 headlines have one content frame, twelve have two, and one has three. The remaining 20 headlines were assigned to the null category as unclassifiable. Economic (15) and Political (14) are the most frequent. There are no positive reference cases for Morality, Fairness and Equality, Crime and Punishment, or Quality of Life.

Against this gold standard, 23 of 70 headlines got no content frame from the model, 42 got at least one, five got two. Micro-averaged precision is 0.77, recall 0.48 and F1 0.59. Macro-averaged figures are lower (precision 0.48, recall 0.34, F1 0.38). Exact-match accuracy is 0.40. Per Label metrics are noted in table 4.

7.2 RSS Extension and Topic Clusters

This extension adds FAZ (Frankfurter Allgemeine Zeitung) and taz (Die Tageszeitung) headlines to Tagesschau’s, all collected September 22, 2026. After deduplication, 281 headlines remain (64 Tagesschau, 167 FAZ, 50 taz), positioned respectively as parliamentary-center, center-rightconservative, and left-liberalprogressive outlets. To approximate a Ground-News-style output, BERTopic clusters the headlines by embedding (Grootendorst, 2022) (paraphrase-multilingual-MiniLM-L12-v2, NLTK German stop words, UMAP for dimensionality reduction, HDBSCAN for cluster-size control), with cosine similaritysilhouette evaluating coherence.

Frame	Precision	Recall	F1
Economic	0.83	0.67	0.74
Capacity and resources	0.50	0.17	0.25
Morality	0.00	0.00	0.00
Fairness and equality	0.00	0.00	0.00
Legality, constitutionality and jurisprudence	1.00	1.00	1.00
Policy prescription and evaluation	0.83	0.62	0.71
Crime and punishment	0.00	0.00	0.00
Security and defense	0.60	0.60	0.60
Health and safety	1.00	0.67	0.80
Quality of life	0.00	0.00	0.00
Cultural identity	0.00	0.00	0.00
Public opinion	0.00	0.00	0.00
Political	0.67	0.71	0.69
External regulation and reputation	0.83	0.62	0.71
Other	1.00	0.10	0.18

Table 4: Per-label metrics. Evaluation of model predictions in comparison with manual gold standard.

Best results: UMAP with 10 neighbors, ten dimensions, min_dist=0.1, seed 42; HDBSCAN minimum cluster size 3, min_samples=5.

This yields 17 topic clusters with relatively high pairwise similarity (mean cosine = 0.73, SD=0.07) but low cosine silhouette (0.14), indicating clusters capture adjacent rather than identical topics grouping similar overarching themes without consistently sharing the same specific event. For example, one cluster may combine election results (opposition focus), weather, and soccer, while a second, more politics-focused cluster revisits the same election with a government-party focus.

Excluding 70 outliers removed during clustering, the saved cluster table has 211 entries. Merging with the model-prediction dataframe leaves 131 data points, with columns for the cluster representative, the cluster’s headlines, predicted frames, and each outlet’s political orientation (Appendix B).

7.3 Interpretation of the Research Questions

RQ1 is partly answered: humans and LLMs recover about 35% and 28%, respectively, of article reference labels at headline level, suggesting headlines convey a recognizable subset of article frames. Because these figures measure reference overlap and can include overlooked frames, they confirm neither every individual headline assignment nor a complete subset relationship between headline and article. As already discussed in Section 4.1, relying on different LLM models and different LLM prompts introduces potential disparities in the annotations that are difficult to trace and therefore complicate comparability of the annotation at a qualitative level. We knowingly accept this trade-off for the sake of expanding the dataset.

For RQ2, fine-tuned ModernGBERT can learn

part of the headline labels despite short inputs, notably on entirely unseen current headlines. Comparing predictions to the gold standard shows high micro- but much lower macro-aggregated metrics, indicating label imbalance: about six of 15 labels score $F1=0$ (Morality, Fairness and Equality, Crime and Punishment, Quality of Life, Cultural Identity, Public Opinion), overlapping with labels that also scored lowest in LLM annotation suggesting both low discriminability and underrepresentation, confirmed by the final dataset’s label distribution (Table 5). Transfer experiments suggest a benefit from adding translated data: CHN+MFC augmentation yields the best macro-F score against German test data versus EU, CHN, or MFC alone; though data volume, topics, and sampling also change, so source-language influence isn’t isolated.

Label	%	Label	%
Quality of life	1.00	Crime and punishment	5.54
Cultural identity	1.29	Economic	6.99
Public opinion	1.39	Security and defense	8.06
Fairness and equality	1.71	External regulation and reputation	9.38
Other	3.16	Political	13.02
Capacity and resources	4.19	Legality, constitutionality and jurisprudence	16.92
Morality	4.54	Policy prescription and evaluation	17.69
Health and safety	5.12		

Table 5: Distribution by label, in percent. Labels marked in bold correspond to those with unstable predictions when applied to current German news headlines.

RQ3 is addressed by the prototype initially at the level of technical feasibility. The application brings together headlines from three outlets, topic clusters, and predicted frames in a single overview. However, because of partially overlapping clusters and the documented selection and linkage issues, a robust comparison of framing for the same event is not yet possible. The qualitative review thus shows both the potential of this comparison and the need for tighter event grouping and a complete presentation of the underlying data.

8 Conclusion

This work¹ examines which article reference frames are visible in headlines, how reliably they

can be classified automatically, and whether this reveals differences between German outlets. Humans and LLMs assign partially overlapping frames but recover only a subset of article references each. LLMs achieve higher precision and internal agreement, humans cover more reference labels justifying our conservative automatic annotation without confirming all headline labels’ correctness.

The qualitative review of the removed labels suggests that headlines often presuppose prior knowledge of people, institutions, and events. Individual words are often insufficient for frame assignment, since their meaning only emerges from context. Translation can introduce additional ambiguity. These observations were not systematically counted as sources of error. Shared model errors and differences between the source corpora are also plausible contributing factors. Culturally driven transfer loss is therefore not demonstrated as an isolated cause.

The fine-tuning results show that ModernG-BERT can learn part of the generated labels. Additional corpora improve overall scores in the shared German transfer test compared with EU data alone. A general benefit of foreign-language data cannot yet be inferred from this, however, since volume, topics, and label distributions change at the same time. In particular, the limited MFC reference and the selective inclusion of null cases shape the training set and its results.

In predicting frames for current German news headlines, the trained model achieves acceptable to good performance for prevalent labels. At the same time, the unstable predictions for all remaining, less prevalent labels reveal the limitations of the data-augmentation process carried out here. Although attempting to predict frames within clustered topics was technically feasible, the limited separation between clusters raises doubts about the meaningfulness of applying label prediction at the level of thematic clusters. Overall, this work thus lays a foundation for further research while also making clear where automatic detection still needs to be validated further. Our contributions include a unified procedure for generating pseudo-labels for headlines from SemEval, MFC, and CHN, together with an annotated resource, a jointly agreed-upon gold standard comprising 70 Tagesschau headlines and a model together with a prototype application for comparing German media².

¹*Declaration of AI Assistance.* Drafted in German by the authors, translated into English and condensed using Claude Sonnet 5, and subsequently co-read, edited, and finalized by the original authors to maintain complete coverage and accuracy.

²*Data and Code Availability.* All code used to conduct the analyses described (data processing pipeline, model training and evaluation scripts), along with all suppl. materials (appendices A and B), is publicly available in the following GitHub repository: github.com/ItsTheTai/frame_detect_NLP.

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